Implementing a Systematic Long-only Momentum Strategy: Evidence From India

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ABSTRACT

We show that a monthly-rebalanced, long-only portfolio of top-decile stocks selected from the NIFTY100 using 'off-the-shelf' momentum criteria significantly outperforms the NIFTY100 Index - both in terms of absolute returns (*by 10.70% pa*) and risk adjusted returns, with a mean turnover of 32.10% per month. We show that momentum persists in the near term but dissipates over time. We demonstrate that our long-only approach has a significant tilt to the momentum factor. We also show that *time in the market rather than timing the market* is important for momentum investing. The strategy has higher volatility and the occasional momentum crash. The strategy's out performance survives real-world implementation given the rise of discount brokers in India. In the absence of cheap ETFs to get exposure to momentum, the systematic long-only strategy from the most liquid part of the market using 'off-the-shelf' criteria provides a practical, executable investment methodology that exposes an investor to momentum in the Indian market.

Keywords— Momentum, Factors, India, Investing, Quantitative, Systematic

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1 Introduction and Literature Review

"far more money is made buying high and selling at even higher prices" - Richard Driehaus

Factor investing is a rational and robust investment approach for long term investors. Factors are investment styles that deliver outsize returns over the long run. Stocks have exposure to **style** factors like value, momentum, and quality. However, factors are not a one way street - they are risky and can under-perform[1]. Equities and bonds are instances of factors where risk premiums are collected by simply buying them and are referred to as *static factors*. Most factor strategies follow Fama and French's (1993) seminal work on factors and build long-short market-neutral portfolios based on a universe of all securities in the relevant market. These *dynamic* factor portfolios are rebalanced regularly. While factor theory is robust, transforming a non-investable risk factor into an investable strategy is complex. Two strategies based on the same factor may behave differently depending upon the choice of implementation (asset universe, choice of metric, cut-off threshold, weighting scheme, rebalancing frequency)[2]. We specifically flag two issues. First, short selling is restricted and often prohibited for retail investors. Alternatives to short selling are expensive and have execution risks not accounted for in theory. Long-only portfolios are *unbalanced* portfolios carrying market exposures as distinct from market-neutral long-short portfolios. Second, market liquidity/capacity is an execution constraint that is difficult to model accurately and practical implementations usually deliver different (and often lower) returns than what theory suggests.

In their original work on buying past winners and selling past losers, Jegadeesh and Titman (1993) laid the foundations of the current momentum framework. They showed picking recent winners from the past 3-12 months and holding them for the next 3-12 months provides abnormal returns. Systemic risks could not explain these returns. Beyond a 12-month holding period, they showed the performance of the winners dissipates over the next 24 months.

Since Jegadeesh and Titman's work, several studies show that the momentum anomaly holds across markets, asset classes and regimes. Rouwenhorst (1998) shows significant momentum returns from a sample of 12 European countries and a sample of 20 emerging markets (1999). In a study of 23 countries across Asia-Pacific, Europe, Japan, and North America, Fama and French (2012) report '*except for Japan, there is return momentum everywhere, and spreads in average momentum returns also decrease from smaller to bigger stocks*'. Asness, Moskowitz, and Pedersen (2013) find evidence for consistent momentum return premia across eight diverse markets and asset classes.

In India, too, momentum studies show evidence of momentum premium. Ansari and Khan (2012) find 'strong presence' of momentum profits in Indian stock between 1995-2006. Maheshwari and Dhankar (2018) using data between 1997-2013 '*confirm the existence of momentum and long-term overreaction effect in the Indian stock market*'.

Any momentum anomaly is difficult to reconcile with the efficient market hypothesis. If the stock price fully reflects all available information, there is no logical explanation as to why stocks which have done well (poorly) in the recent past will continue to do well (poorly) in the 'near term'. Explanations for the anomaly can be categorised into two broad themes:

- **Economic**: the higher returns compensate the additional risk investors take when they invest in a momentum strategy (see Li, Miffre, Brooks, and O'Sullivan, 2008).

[1] occasionally for extended periods: the value factor, for instance, has been under-performing for over a decade!

[2] Israel, Jiang, and Ross (2017a) explore the targeting and capture of factor premia and suggest that execution and implementation choices may constitute a form of alpha which they call '*craftmanship alpha*'.

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- **Behavioural**: the price under- and over-reaction results from the cognitive errors investors make when incorporating information. For instance, investors have a *conservatism bias* - they are slow to adjust their investment views when presented with new (and especially contradictory) information. Or investors use the *representativeness heuristic* - they are too quick to conclude that a stock follows a particular 'ideal type'. Citing evidence that over 3- to 5- year holding periods, stocks which performed poorly over the previous 3 to 5 years had higher returns than stocks that performed well over the same period, Bondt and Thaler (1985, 1987) argued that investors tend to overreact to information. Daniel, Hirshleifer, and Subrahmanyam (1998) build a behavioural model based on the overconfidence of individuals to explain the momentum anomaly. Barberis, Shleifer, and Vishny (1998) propose a model that is consistent with the empirical findings of pervasive *under reaction* of stock prices to news such as earnings announcements, and *over reaction* of stock prices to a series of good (bad) news. The behavioural attributes create momentum as stock prices react with delay to firm-specific information and subsequent reversion once deviations from equilibrium are acknowledged.

Despite widespread empirical evidence[3] for the momentum anomaly and some strong theoretical hypothesis to explain it, there remain serious debates around momentum investing. The most significant is whether the out-performance of such portfolios survive transaction costs in the real-world. The intuitive economic argument is simple. Investors will demand more of the stocks selected by the strategy creating upward price pressure that will, inevitably, lower expected returns. Conversely, when investors exit existing positions, their selling lowers prices, reducing realized return. Unfortunately, these dynamics of demand and supply on capacity and costs are not easily observable.

A number of studies have estimated capacity for factor-strategies (see Zhiwu Chen (2002), Andrea Frazzini (2012), Novy-Marx and Velikov (2014) and Ratcliffe et al. (2017)). Vogel (2017) surveys literature on both sides of the survival of momentum returns in the real world and concludes that the answer to the cost/capacity question depends on the assumptions and data used. Feifei, Tzee-Man, Pickard, and Garg (2019) build a model for the market impact of such costs for a number of factor-based strategies, including momentum. Israel, Ross, Moskowitz, and Serban (2017b) use the live track record of the implementation of momentum by AQR and '*show that momentum is an implementable strategy and that the live experience of trading has not necessarily produced excessive turnover, large trading costs, or significant tax burdens. These results contradict common myths about momentum investing*'. Frazzini, Israel, and Moskowitz (2018) using data from actual trading costs from a large institutional asset manager argue that actual costs from live trades are significantly lower than what academic studies suggest.

Despite the ongoing debate on the survival of momentum returns in the real-world, there is unanimous consensus that excessive transaction costs are a prime destroyer of excess returns. Novy-Marx and Velikov (2014) show that most of the factors with one-sided monthly turnover higher than 50% are not profitable after trading costs. Transaction costs reduce the factor profitability and its associated statistical significance. For an investor, transaction costs are primarily brokerage, custody and taxes. Other than taxes, costs are being driven down by the increasing use of fintech, digitisation and digitalisation. Portfolio turnover remains an important determinant of whether a practical implementation is profitable.

Unfortunately, similar studies on momentum cost/capacity do not exist for the Indian market. Kumar and Misra (2019) find '*liquidity forms part of the systematic and idiosyncratic risk. Therefore, failure to incorporate it into portfolio formulation strategies may lead investors on NSE to take er-*

[3] across time, asset classes and geographies

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roneous investment decisions'. Given these conceptual liquidity considerations, we choose to restrict our universe to the NIFTY100 stocks, thereby operating in the most liquid area of the market. Most retail trades are likely to be a small fraction of the overall daily volume traded for NIFTY100 constituents. Momentum strategies, therefore, should not suffer because of capacity constraints. Unlike the US, there are no publicly traded momentum exchange-traded funds (ETFs) in the Indian market. In theory, therefore, momentum stocks currently have little systematic demand.

A word about long-short portfolios. These portfolios are long the preferred exposure and short the unwanted exposure in equal measure. For exposure to the momentum factor, the portfolio would buy stocks exhibiting strong positive momentum and sell stocks with strong negative momentum. In reality, it is far easier to buy stocks than it is to sell stocks that one does not own. The short leg is challenging at best and often impossible to execute. It should be no surprise that many of the factor ETFs listed in the US are actually long-only portfolios that are *tilted* toward or away from particular factor exposures. Long-only factor portfolios will not fully replicate the performance of long-short factors. Huij, Lansdorp, Blitz, and van Vliet (2014) examine the differences between long-only and long-short portfolios and find '*that although a long-short approach is superior theoretically, a long-only approach seems to be the preferred alternative in most scenarios, after accounting for practical issues such as benchmark restrictions, implementation costs and factor decay*'[4].

Accessing factors cheaply is impossible for most non-institutional investors in India. There are no ETFs offering specific factor exposure and neither do mutual funds give such targeted exposure. For a self-directed investor, any Do-It-Yourself factor-exposure portfolio would need to be easy to administer and cheap to execute. With the growth of discount brokers, transaction costs, like brokerage, have dropped significantly for all investors. Structured data is increasingly accessible at a reasonable cost. Retail investors can easily access and use data from financial reports and price data from exchanges to build and manage their investment portfolios.

Given the constraints of short selling in India, the lack low-cost instruments that target factors, the low cost of brokerage and the increasingly easy access to structured data it seems appropriate, therefore, to examine the possibility of building a systematic long-only momentum portfolio. We look to build an easily understood and practically implementable strategy using 'off-the-academic-shelf' measures. Our objective is to robustly examine the practical implementation of such a strategy. We recognize that there can be different implementation choices leading to different outcomes. We do not seek to optimize a strategy using data analysis and falling prey to data snooping[5] and/or over-fitting.

This paper attempts to translate factor theory into practice for the general self-directed investor and extends current thinking in at least three areas. First, geographically, we believe it is one of the early in-depth exploration of build practical momentum strategies within the Indian context. Second, quantitative, systematic strategies are still largely of interest amongst institutional investors. This paper is one of the early efforts to extend factor theory to non-institutional investors. We believe both areas require more research and debate to raise the standards of discussion amongst and choices of investment for the wider investment community. Third, the paper makes an essential distinction between the signal for portfolio construction and the execution price for rebalancing. Traditionally, most studies have not made this distinction, and we are one of the early papers to explore this aspect within an Indian context.

The rest of the paper is set out as follows: Section 2 outlines the sample and data we use. Section

[4]See also Cazalet and Roncalli (2014)

[5]Data snooping refers to statistical inference performed by a researcher after looking at the data.

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3 is the main body of the paper which has a series of subsections: Section 3.1 defines our momentum measure; Section 3.2 decomposes momentum returns in more detail, specifically looking into the effects of long-only and size of firm; Section 3.3 examines our choice of the NIFTY100 index for any inherent biases using Fama French Factors; Section 3.4 examines the question of overnight risk for our implementation; Section 3.5 examines if momentum is persistent; Section 3.6 finally, analyses the returns of our strategy. Section 4 concludes.

2 Sample and Data

All data is from Refinitiv. Price data is daily close prices for the purposes of analysis and portfolio construction and daily open prices for investment/rebalancing. For stocks that have been delisted, we rely on price data available on Refinitiv[6]. All returns are price returns in INR unless stated otherwise.

Our universe is the constituents stocks of the NIFTY100 Index (NIFTY100) during the period December 2010 through September 2019. NIFTY100 is a diversified ≈100 stock index representing major sectors of the Indian economy representing the (approximately) top 100 companies[7] based on full market capitalisation from NIFTY 500. This index tracks the combined portfolios of the NIFTY50 and the NIFTY NEXT 50 indices and measures the performance of large market capitalisation companies (see Nifty Indices, 2019). The NIFTY 100 Index represents about 76.8% of the free float market capitalization of the stocks listed on NSE as on March 29, 2019. The total traded value for the last six months ending March 2019 of all index constituents is approximately 66.2% of the traded value of all stocks on the NSE (NSE India Website). The index is re-balanced semi-annually. We use the NIFTY100 Index constituents every month end.

We use survivorship-bias adjusted data from IIM, Ahmedabad's factor database (Agarwalla, Jacob, and Varma, 2013) for our analysis. The database runs from October 1993 to September 2019. In our analysis of the momentum factor we use this period. However, when we analyse the strategy or NIFTY100, we use the period from January 2010 to September 2019.

We compute our first portfolio using data as of close on the last day of trading in December 2010 - the first portfolio is created at open on the first trading day in January 2011. Our last portfolio analysis date is the last trading day of August 2019 (portfolio created on the first trading day of September 2019).

3 Analysis

3.1 Defining Momentum

Momentum is commonly defined as the last 12 months of returns skipping the last month[8]. We define momentum for stock i as Equation 1

$$r_t^i = log\left(\frac{P^i_{close_{t-1}}}{P^i_{close_{t-12}}}\right) \tag{1}$$

where r_t^i is the momentum price return for stock i for month t, $P_{close_{t-1}}$ is the closing price for stock i on the last trading day of month $t-1$ and $P^i_{close_{t-12}}$ is the closing price for stock i on the last trading

[6]In most cases, the data is available. On the occasional instance where data is not available, we ignore the stock

[7]The NIFTY100 has 101 constituents as of end September 2019

[8]Jegadeesh and Titman (1993) and many other researchers also use a skip period (ranging from a week to a month) to avoid issues related to bid-ask spread, price pressure and lagged reaction effects.

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day month $t-12$.

Following Jegadeesh and Titman (1993) nomenclature, 11[9] months is the formation period (J) and the holding period (K) is for 1 calendar month for the strategy. At the end of the last trading day of each calendar month, we rank the returns based on closing prices for all the constituent stocks of the NIFTY100 for the month and choose a decile portfolio of stocks with the largest returns. These stocks are then rebalanced at the open price of the next trading day[10] (the first trading day of the following month).

We choose a **top decile portfolio**. This is easy to execute. Managing a 10 stock portfolio requires little overhead by an investor in terms of administration and management. It is, however, a concentrated portfolio and may not be appropriate for all investors. Investors should choose an appropriate number of stocks based on their individual risk appetite and loss aversion preferences.

We choose an **equal weight portfolio**. This is intuitive and convenient. Other approaches to portfolio construction include mean-variance optimization, market cap weights.[11]

We rebalance at the **market open on first trading day of every month**: it is systematic and convenient[12].

Our approach is different from the approach taken by Agarwalla, Jacob, and Varma (2013) in the construction of the Fama-French factors. We flag the differences:

- Agarwalla et al. include in their universe all firms listed on the Bombay Stock Exchange (BSE) covered in the CMIE Prowess database between 1991-2011[13] with valid prices and meet a liquidity filter. They cover a total of 6,943 firms before their filters take effect. Agarwalla et al. report that the median number of firms they categorize into Big[14] size momentum portfolios was 136.

- Agarwalla et al. define momentum returns, like us, at the end of month t as the 11- month returns from the end of month $t-12$ to $t-1$. However, they use *total returns (including dividends)* at the end of each month to calculate momentum. We use price momentum. This difference could affect outcomes in at least two ways. First, the portfolio selected could change month on month depending on the payment of dividends. Second, if the market is not efficient, dividend information/expectations would not be fully reflected in the current price of the stock leading to differences in portfolio selection. We believe that the differences are minor and therefore do not affect the overall results.

  Agarwalla et al. estimate monthly returns even when a stock is traded only for a part of its first month of trading using the available daily returns. We ignore stocks that do not have 252 days of trading history. It is difficult to quantify this impact, but we believe that the difference, if any, will be negligible.

- Agarwalla et al. group stocks using total momentum returns into two groups:

[9] 12-months-skip-recent-month

[10] This is a pragmatic and practical execution strategy - most online brokerages allow investors to place 'at-market' or 'without limit' overnight orders which get placed into the pre-opening session of the exchange for execution. As the stocks in the NIFTY100 are liquid, the execution invariably happens at the open prices.

[11] We do not consider the various weighting alternatives in this paper.

[12] Calendar effects have been studied in many markets. (see, for instance Ariel, 1987)

[13] The factor database is updated every month and so their universe remains current.

[14] Firms were classified into Big (B) and Small (S) based on their market capitalization at the end of September each year. See table below.

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– a Winner (W) group consisting of the top 30% by the momentum return

– a Loser (L) group consisting of the bottom 30% by the momentum return.

We use deciles to construct our 'winner' group resulting in a more concentrated portfolio. *A priori*, we would expect our more concentrated portfolio to potentially have higher volatility than the Agarwalla et al. factor portfolio.

Following Fama and French (2012), Agarwalla et al. orthogonalize their momentum portfolio to size. They create four size and momentum portfolios - WS, WB, LB and LB - every month (Table 1) and define '*big firms (B) as the top 10% by market capitalization and classified the remaining firms as small firms (S)*'. Further, Agarwalla et al. report that '*The Indian market was dominated by a large number of small firms. For instance, the market capitalization of the 90th percentile firm was around ₹0.7 billion in 1997, ₹7 billion in 2004 and ₹16 billion in 2012*'. The NIFTY100 stocks are well in excess of the 90th percentile cut-off.

Table 1: Size-Momentum Portfolios from Agarwalla et al.

		Momentum	
		Winner (W)	Loser (L)
Size	Big (B)	WB	LB
	Small (S)	WS	LS

Table 2 reports the median number of firms in the size-momentum buckets in Agarwalla et al. (2013). For months when the number of stocks in any bucket were less than 5 they exclude the relevant sub-portfolio.

Table 2: Size-Momentum Portfolios from Agarwalla et al.

		Momentum	
		Winner (W)	Loser (L)
Size	Big (B)	105	31
	Small (S)	669	726

Our portfolio drawn from NIFTY100 will lie entirely in the Big (B) category.

3.2 Decomposing momentum within the Indian context

The momentum factor (WML[15]) is calculated by Agarwalla et al. (2013) following Fama and French (2012) as a simple average largely free of the influence of size of the firm (Equation 2).

$$WML = \frac{(WB - LB) + (WS - LS)}{2} \tag{2}$$

Does momentum beat the market? To explore this, we can use the traditional method of examining excess return with respect to the CAPM one-factor model; and the Fama and French (1993) three-factor model, which includes the market factor MKT, the size factor SMB[16] and the value factor

[15] Winner-Minus-Loser is also referred to as UMD Up-Minus-Down. We will follow the WML nomenclature in this paper.

[16] Small-minus-Big

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HML[17]. The WML returns are treated as those from the investor's portfolio. We look specifically for the intercept : a positive intercept reflects out-performance (more traditionally, α) in equation 3 with the first one or three right hand-side variables included:

$$WML_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \epsilon_t \tag{3}$$

where WML_t is the return of the momentum factor for period t, MKT_t is the Market excess return for period t over the risk free rate for period t and SMB and HML are the Small-minus-Big (size) and High-minus-Low (value) factors for period t respectively. Note as WML is a market-neutral 'portfolio', we do not calculate excess returns. Table 3 shows the results. Specifically, there is a significant positive intercept (CAPM: 22.74% and FF3: 23.85% annually), and the market returns are poorly correlated with WML returns. Of note, WML is negatively loaded to HML: momentum is inversely related to value (see, for instance Asness et al., 2013). And, the R^2 is expectedly low since WML is market neutral.

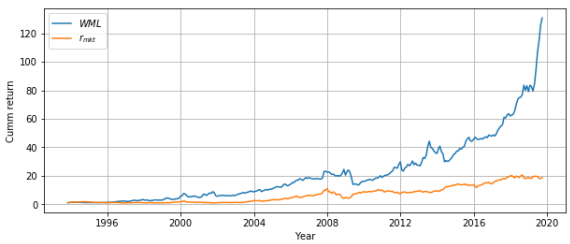
Table 3: WML : CAPM and FF3 Regressions

	CAPM			FF3		
	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue
Intercept	1.90	4.59	0.00	1.99	4.83	0.00
MKT	-0.16	-2.85	0.00	-0.12	-2.11	0.04
SMB				-0.02	-0.22	0.82
HML				-0.16	-2.42	0.02
Adjusted R2	0.02		0.00	0.04		0.00

This table shows calendar-time portfolio returns. The dependent variable is the monthly returns of the momentum factor WML, the explanatory variables are excess returns of market (MKT), size (SMB) and book-to-market (HML) portfolios from Agarwalla et al. (2013). The sample period runs from Oct 1993 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates.

The cumulative returns from WML and the market portfolios (r_{mkt}) are shown in Figure 1. Table 4 shows the descriptive statistics. The data is from Agarwalla et al. (2013). The WML monthly distribution has a positive skew and a better raw sharpe than the market[18]. The evidence for the momentum anomaly is clear in the Indian market.

Figure 1: Market and WML monthly returns



Note: monthly returns from survivorship-bias adjusted data. The cumulative returns are cumulative product of monthly returns from October 1993 to September 2019

Table 4: Monthly Returns Market and WML

	Market	WML
nobs	312.0000	312.0000
min	-0.2812	-0.2605
max	0.3576	0.3450
mean	0.0123	0.0179
std dev	0.0756	0.0735
skewness	-0.0362	0.1017
kurtosis	1.8147	3.0040
raw Sharpe (ann)	0.5633	0.8456

This table shows descriptive statistics of the monthly returns of the Market and WML from Agarwalla et al. (2013). We compute the monthly returns for WML from daily returns.

Next, we examine both axes of the size-momentum universe. Tables 5 and 6 report the descriptive statistics for the monthly returns for WML and its four size momentum portfolios (from Table 1 : WB, WS, LB and LS) to end September 2019 both from inception and past 60 months. The contribution of the *Winner* portfolios (long only) is evident - the returns (means), risk-adjusted returns (raw sharpes) are far superior to the *Loser* portfolios (short only). Within *Winner*, the size effect

[17] High-minus-Low

[18] Note that the market is derived from Agarwalla et al. (2013) universe rather than any public benchmark.

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is more nuanced - both WB and WS have similar raw sharpes since inception. WS is a platykurtic distribution with a left tail similar to the less platykurtic WB since inception. More recently (past 60 months), WB has the higher raw sharpe and is less left-skewed than WS. The *Loser* portfolios are, in general, similar to each other - lower mean returns, platykurtic and longer left-tails than their *Winner* counterparts. Over the entire period, the *Loser* portfolios have detracted from WML performance, although in the past 60 months they have been strong contributors.

Figures 2 and 3 show the histograms of monthly returns for WML and its components. The left tail risks of returns from these portfolios compared to the returns of WML portfolio are clearly visible. These left tail risks have been called "*momentum crashes*" by Daniel and Moskowitz (2016). The *Loser* portfolios, irrespective of size, have longer left tails than their *Winner* counterparts. Within the Indian equities context, *going short gives lower mean returns and has greater left-tail risk.*

Table 5: Monthly Returns WML and components: since inception

	WB	WS	LB	LS	WML
nobs	312.0000	312.0000	312.0000	312.0000	312.0000
min	-0.2983	-0.3057	-0.6014	-0.6972	-0.2605
max	0.3865	0.3601	0.3965	0.3375	0.3450
mean	0.0186	0.0200	-0.0047	-0.0034	0.0179
std dev	0.0859	0.0931	0.1159	0.1134	0.0735
skewness	0.3042	0.0018	-0.5561	-1.0154	0.1017
kurtosis	2.7433	0.9851	3.2323	4.7533	3.0040
raw Sharpe (ann)	0.7512	0.7448	-0.1417	-0.1045	0.8456

This table shows descriptive statistics of the components of WML. We convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series. The LB and LS signs have been inverted.

Table 6: Monthly Returns WML and components: past 60 months

	WB	WS	LB	LS	WML
nobs	60.0000	60.0000	60.0000	60.0000	60.0000
min	-0.0846	-0.1863	-0.1644	-0.1131	-0.0555
max	0.1176	0.1428	0.1901	0.1713	0.1291
mean	0.0148	0.0130	0.0106	0.0094	0.0246
std dev	0.0460	0.0675	0.0684	0.0694	0.0389
skewness	0.0052	-0.4884	0.2941	0.3947	0.4174
kurtosis	-0.4357	0.4244	0.3179	-0.5600	0.2836
raw Sharpe (ann)	1.1149	0.6658	0.5366	0.4666	2.1891

This table shows descriptive statistics of the components of WML. We convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series. The LB and LS signs have been inverted.

To examine whether the distributions are similar, we run Kolmogorov-Smirnov (K-S) two sample tests on the monthly return series since inception for each pair of WML and its components. Tables 7 and 8 shows the cross-tables of the Kolmogorov-Smirnov statistics and their associated p-values[19] for the pair-wise tests for the full time period. For WML and WB, the p-value is significant and therefore implies that the two distributions are similar. The two winner portfolios differentiated by size (WB and WS), have different return distributions given their p-value. The skewness and kurtosis of the two distributions are significantly different. *Ceterius paribus*, having a mix of WB and WS would be preferred to having only WB in a portfolio given the diversification benefits. However, in the Indian context, small market cap firms have very different risk profiles than large cap firms including liquidity/capacity and reporting and governance standards. These considerations, often difficult to quantify, are important in portfolio construction decisions. In our choice of the NIFTY100 as the universe, we do not include any *Small* companies by construction.

Table 7: Kolmogorov-Smirnov statistic D for WML and components: since inception

	WB	WS	LB	LS	WML
WB	0.0000	0.0994	0.1891	0.1538	0.0513
WS	0.0994	0.0000	0.1410	0.1314	0.1154
LB	0.1891	0.1410	0.0000	0.0641	0.2019
LS	0.1538	0.1314	0.0641	0.0000	0.1795
WML	0.0513	0.1154	0.2019	0.1795	0.0000

This table shows the $D_{n,m}$ where n and m are a pair from WB, WS, LB, LS, and WML

Table 8: p-value of Kolmogorov-Smirnov statistic D for WML and components: since inception

	WB	WS	LB	LS	WML
WB	1.0000	0.0919	0.0000	0.0012	0.8074
WS	0.0919	1.0000	0.0040	0.0091	0.0313
LB	0.0000	0.0040	1.0000	0.5438	0.0000
LS	0.0012	0.0091	0.5438	1.0000	0.0001
WML	0.8074	0.0313	0.0000	0.0001	1.0000

This table shows the p-value for the KS statistic $D_{n,m}$ where n and m are a pair from WB, WS, LB, LS, and WML

Both the *Loser* portfolios, LB and LS, have significant p-values indicating that both return distributions are similar to each other and different to WML return distribution. Figure 2 shows the monthly returns distributions from inception graphically.

[19] For large sample sizes, the approximate critical value D_α is given by $D_\alpha = c(\alpha)\sqrt{\frac{n+m}{nm}}$ where $c(\alpha)$ is the coefficient from the KS Two Sample table and n, m are sample sizes ($n = m = 312$). $c(\alpha)$ at $1\% = 1.63$ and at $5\% = 1.36$, so the D_α will be 0.1305 (1%) or 0.1089 (5%)

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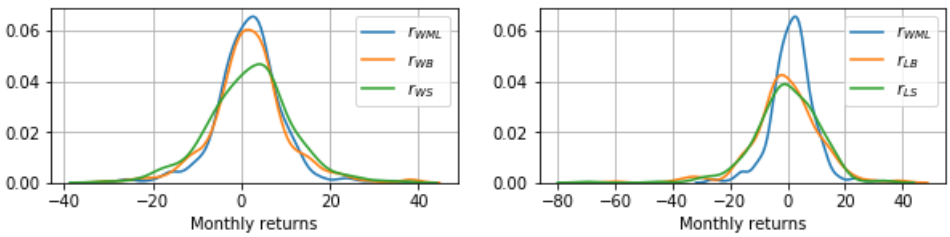
Figure 2: Distribution of Monthly Returns:WML and components : since inception



We convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

Next, to examine the more recent period, we do the same analysis for the past 60 months[20] in Tables 9 and 10. This is a more noisy period as evident from Figure 3. All of the components have longer left tails and are more playtkurtic than historical averages. With this context, key points of note: First, WS-WML and WS-WB have significant p-values. However, WB and WML do not. Therefore, the logical conclusion that all three return distributions are similar fails[21]. WS-WB have a higher K-S p-value indicating that these two return distributions are closer. In the face of this conflicting result, we conclude that WB and WS have similar distributions but neither of the two is similar to WML. LS and LB continue to show similar distributions. The *Loser* portfolio has contributed to WML performance in this period. In both the longer sample and the truncated sample, LB and LS return distributions are similar and remain different from WML return distribution. The long left tails of the loser portfolios remain irrespective of regime. The *Winner* portfolio is a significant driver of WML, while the *Loser* portfolio in general is a detractor, with occasional high contribution to WML.

Table 9: Kolmogorov-Smirnov statistic D for WML and components: past 60 months

	WB	WS	LB	LS	WML
WB	0.0000	0.1167	0.1500	0.2000	0.2000
WS	0.1167	0.0000	0.1333	0.1667	0.1667
LB	0.1500	0.1333	0.0000	0.1000	0.2833
LS	0.2000	0.1667	0.1000	0.0000	0.3000
WML	0.2000	0.1667	0.2833	0.3000	0.0000

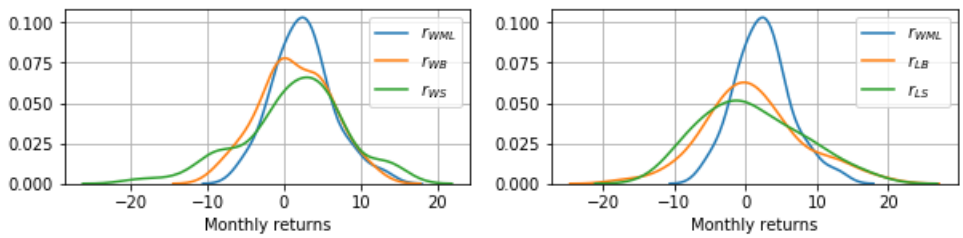
This table shows the $D_{n,m}$ where n and m are a pair from WB, WS, LB, LS, and WML

Table 10: p-value of Kolmogorov-Smirnov statistic D for WML and components: past 60 months

	WB	WS	LB	LS	WML
WB	1.0000	0.8133	0.5130	0.1821	0.1821
WS	0.8133	1.0000	0.6648	0.3777	0.3777
LB	0.5130	0.6648	1.0000	0.9284	0.0158
LS	0.1821	0.3777	0.9284	1.0000	0.0087
WML	0.1821	0.3777	0.0158	0.0087	1.0000

This table shows the p-value for the KS statistic $D_{n,m}$ where n and m are a pair from WB, WS, LB, LS, and WML

Figure 3: Distribution of Monthly Returns:WML and components : past 60 months



We convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

To explore the interactions further, we break down WML by Size (B and S) and then Momentum (W and L). For Size, Equation 2 can be re-written as :

$$WML = \frac{1}{2}WML_{Big} + \frac{1}{2}WML_{Small} \tag{4}$$

[20]Note p-values change to 0.2976 (1%) and 0.2483 (5%), as the sample size is now 60

[21]The logical statement : If $A = B$ and $B = C$, then $A = C$.

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where WML_{Big} and WML_{Small} denote the returns from momentum portfolios created by size - *Big* and *Small* - respectively. Figure 4 shows the cumulative monthly returns since inception. While both *Big* and *Small* portfolios contribute to WML, *Small* clearly contributes significantly more across the entire period. This is line with evidence that *Small* firms demonstrate higher momentum effects. India being a growth economy, this result seems intuitive and aligned to the Size factor premium evidenced in the Indian market (see, for instance, Connor and Sehgal, 2001). Over the past 60 months (Figure 5), however, the contribution weights of *Big* and *Small* becomes more even. In 2019, *Big* contributes more to WML than *Small*. This, we hypothesise, is a result of the increased headwinds in the Indian economy - slowdown of economy and credit - leading to a flight to quality (the large cap, well established firms).

Figure 4: Indexed WML, WML_{Big} and WML_{Small} since inception

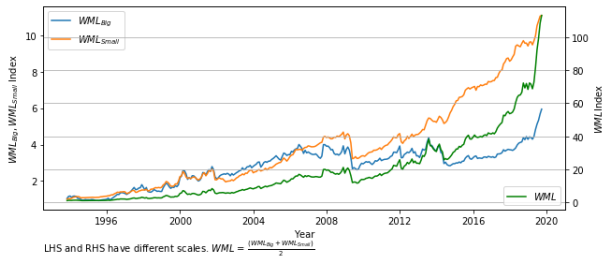
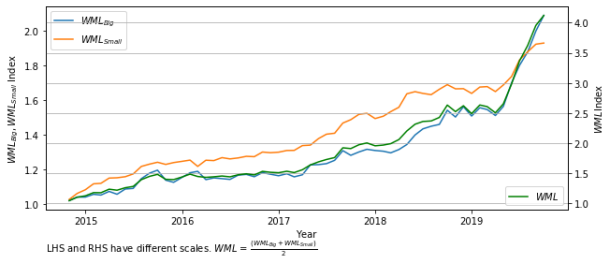


Figure 5: Indexed WML, WML_{Big} and WML_{Small} past 60 months



Note: monthly returns from survivorship-bias adjusted data. The cumulative returns are cumulative product of monthly returns from to September 2019. For WB we convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

Next, we look at Momentum. What is the contribution to WML from the Long ($Winner$) or Short ($Loser$) legs? We can re-write Equation 2 as:

$$\begin{aligned} WML &= \frac{(WB + WS) - (LB + LS)}{2} \\ &= \frac{1}{2}Winner - \frac{1}{2}Loser \end{aligned} \tag{5}$$

where $Winner$, the winners, reflect the returns of the longs and $Loser$, the losers, reflects the returns of the shorts. Figures 6 and 7 graphically illustrate cumulative returns for W and L across the two periods. Since, inception $Winner$ has dominated. This is similar to results in other markets (see Huij et al., 2014; Griffin et al., 2004, amongst others). India's growth bias is well reflected in this data - *in a growing economy, buying growth has better returns than going short growth.* Since 2018, however, $Loser$ has contributed more to WML than historical mean contributions. This reflects, we argue, the current conditions of the economy and the equity market, with large swings to the downside. Once the growth trajectory is re-established, the contributions from $Winner$ and $Loser$ will, we posit, revert to mean.

Next, we isolate the cumulative return of the Size components of $Winner$ in Figures 8 and 9 to show their contribution ($Winner_{Big}$ and $Winner_{Small}$ respectively). Both WB and WS do contribute to WML. As we said before, there is clearly merit to adding *Small* into a momentum strategy depending on the risk appetite/tolerance of the investor. Since 2018, WS has been a net detractor to WML's performance - reflecting the flight to quality in the face of the recent economic conditions.

Having understood in more detail the drivers for WML, importantly, given our construction choice of the NIFTY100 as our universe, does WB also outperform the market? Figure 10 and 11 shows

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Figure 6: Indexed WML, W and L since inception

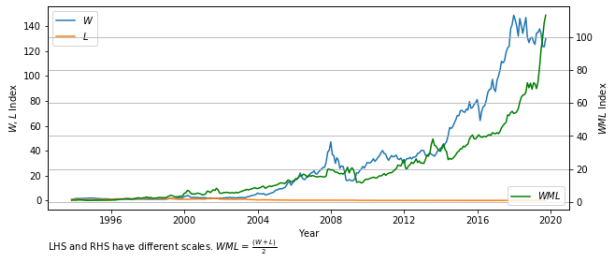
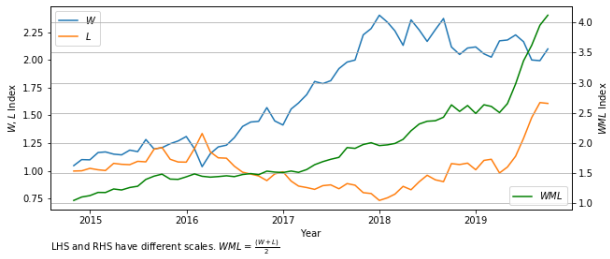


Figure 7: Indexed WML, W and L past 60 months



Note: monthly returns from survivorship-bias adjusted data. The cumulative returns are cumulative product of monthly returns from to September 2019. For WB we convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

Figure 8: Indexed WML, WB and WS since inception

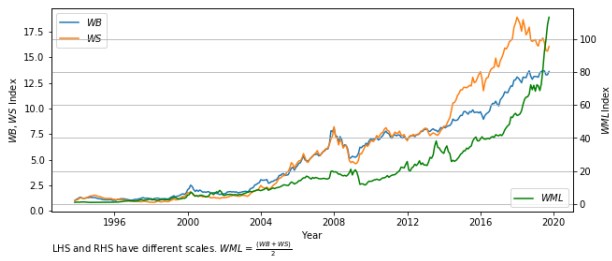
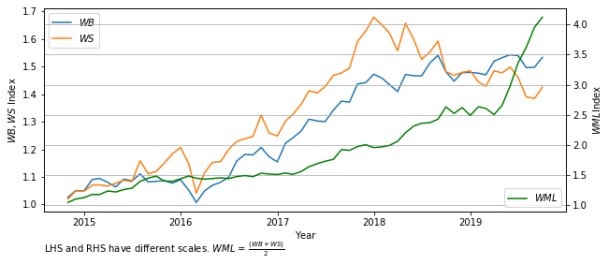


Figure 9: Indexed WML, WB and WS past 60 months



Note: monthly returns from survivorship-bias adjusted data. The cumulative returns are cumulative product of monthly returns from to September 2019. For WB we convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

the trend of the cumulative returns of both return distributions. More formally, we examine excess returns of WB portfolio and look at the intercept (α) with respect to the CAPM one-factor model; the Fama and French (1993) three-factor model (FF3), which includes MKT, SMB and HML; and extending the Fama-French three-factor model to include WML (FF4) (Jegadeesh and Titman (1993), Carhart (1997), Fama and French (2011)). The intercepts are from the regression equation 6 with the first one, three or four right hand-side variables included:

$$[WB_t - RF_t] = \alpha + \beta_{MKT}MKT_t + \beta_{SMB}SMB_t + \beta_{HML}HML_t + \beta_{WML}WML_t + \epsilon_t \tag{6}$$

where WB_t is the return of the WB portfolio for period t, MKT_t is the Market excess return for period t, RF_t is the risk free rate for period t and SMB, HML and WML the size, value and momentum factors for period t. Ss WB has market exposure, we calculate excess returns. Table 11 shows the results. Specifically, there is a significant positive alpha (CAPM: 7.34% and FF3: 7.75% annually) against CAPM and FF3. The α is less than that of WML as expected - but WB outperforms the market. WB is also negatively loaded to HML, though weaker than the market-neutral WML. Importantly, WB has a positive and significant coefficient to WML - the positive tilt to momentum is clear and evident with the WML. Further, the adjusted R^2 with FF4 shows a significant jump from FF3 - the relationship between WML and WB is clear.

As our universe is limited to *Big* and we are long only, the contributions, mainly from WS and occasionally from *Loser* components are missing. By excluding WS, a significant source of WML performance is being ignored. However, we argue, the WML returns are theoretical and will not survive a practical implementation. First, *Loser* portfolios are not possible to execute for the general investor. So the contribution will not translate in real portfolios. Executing shorts through futures is restrictive (as only a few stocks, mainly *Big* have liquid futures) and the costs will reduce the returns. Second, $Winner_{Small}$ returns are likely to not survive liquidity related slippages. Therefore, realized $Winner_{Small}$ and *Loser* portfolio returns are likely to be significantly different from what Agarwalla

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Table 11: WB: CAPM, FF3, FF4 Regressions

	CAPM			FF3			FF4		
	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue
Intercept	0.61	2.97	0.00	0.65	3.14	0.00	-0.17	-1.33	0.18
MKT	1.03	38.09	0.00	1.05	37.45	0.00	1.10	64.76	0.00
SMB				-0.01	-0.13	0.90	0.00	0.05	0.96
HML				-0.06	-1.81	0.07	0.01	0.24	0.81
WML							0.40	23.44	0.00
Adjusted R2	0.82		0.00	0.82		0.00	0.94		0.00

This table shows calendar-time portfolio returns. The dependent variable is the monthly excess returns of the WB portfolio, the explanatory variables are excess returns of market (MKT), size (SMB) and book-to-market (HML) portfolios from Agarwalla et al. (2013). The sample period runs from Oct 1993 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates.

et al. calculate. $Winner_{Big}$ portfolio returns are more translatable into a real strategy given the liquidity associated with the stocks. Importantly, $Winner_{Big}$ outperforming the market shows that a momentum tilt could form the basis for an investor to translate a non-investable factor to an investable strategy and gain exposure to the momentum factor.

Figure 10: Market, WML, WB Monthly Returns From Inception

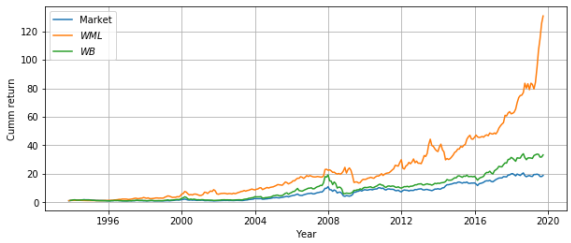
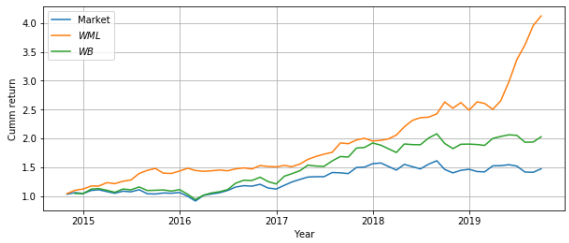


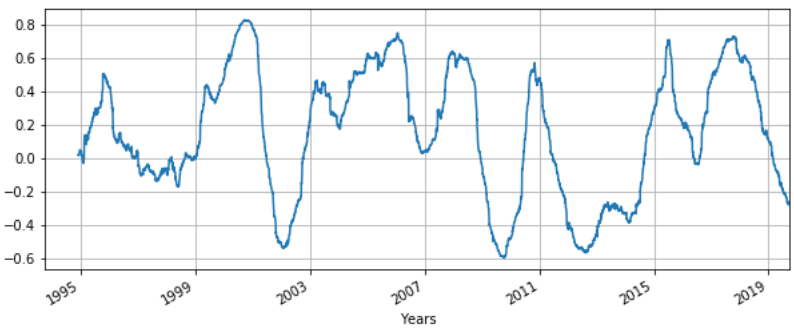
Figure 11: Market, WML, WB Monthly Returns past 60 months



Note: monthly returns from survivorship-bias adjusted data. The cumulative returns are cumulative product of monthly returns from to September 2019. For WB we convert daily returns from survivorship-bias adjusted data (Agarwalla et al., 2013) to a monthly series using cumulative product to build the time series

One final point: the correlation between WB and WML. Figure 12 shows the rolling 252 day correlations between the daily returns of the two. Note the rather interesting cyclical trend of this correlation. What can be seen from the chart is the impact the composite WS and *Loser* portfolios have to WML. The correlation breaks down or reverses as we get into certain economic regimes when the returns of the short side increase (economic stress) or returns from the more nimble WS (parts of an economic expansion) contribute significantly to WML. More work is needed to understand this trend better.

Figure 12: 252 day Rolling Correlations between WML and WB



In conclusion, decomposing WML into the longs and shorts and by size has shown that the $Winner$ portfolio (the long portfolio) contributes more than the *Loser* portfolio (the shorts). Theoretically, the $Winner_{Small}$ portfolio contributes significantly although the contribution in recent years has fallen

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from long term averages. However, these returns are unlikely to survive a real implementation because of liquidity slippages. *A long only momentum tilt should be attractive* to an investor seeking to outperform the market over time. Our choice of NIFTY100 as the universe for constructing a long-only momentum tilt portfolio seems appropriate.

3.3 NIFTY100 and Market

Agarwalla et al. (2013) calculate market returns from a significantly larger base of stocks than the NIFTY100. We need to ensure that the choice of our universe does not have an inherent momentum bias. To explore this, we examine excess return with respect to the CAPM, FF3 and FF4 models. The alphas are the intercepts from the regression (Equation 7) with the first one, three or four right-hand-side variables included:

$$[r_{NIFTY100_t} - RF_t] = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{WML} WML_t + \epsilon_t \quad (7)$$

where $r_{NIFTY100_t}$ is the monthly return of NIFTY100 for period t, RF_t the risk-free rate for period t, MKT_t is the Market excess returns and SMB_t, HML_t and WML_t are the respective factor returns for period t, α, the intercept and β being the respective betas for period t. For the various factors in Equation 7, we use the dataset from Agarwalla et al. (2013)[22]. Our analysis period is January 2011 through September 2019. Table 12 shows the results.

Table 12: NIFTY100: CAPM, FF3 AND FF4 Regressions

	CAPM			FF3			FF4		
	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue
Intercept	-0.08	-0.65	0.52	-0.10	-1.14	0.26	-0.13	-1.25	0.22
MKT	0.98	37.59	0.00	1.03	43.73	0.00	1.04	41.04	0.00
SMB				-0.15	-5.84	0.00	-0.15	-5.84	0.00
HML				-0.09	-4.71	0.00	-0.09	-4.53	0.00
WML							0.01	0.52	0.60
Adjusted R2	0.95		0.00	0.97		0.00	0.97		0.00

This table shows monthly returns. The dependent variable is the monthly excess returns of NIFTY100 using closing prices. The explanatory variables are the returns of the market (MKT), size (SMB), book-to-market (HML), and momentum (WML) portfolios from Agarwalla et al. (2013). The sample period runs from Jan 2011 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates.

The CAPM β of the NIFTY100 is 0.981. We expect the β would be slightly different from 1 given the differences in the universes so this minor variation is unsurprising. Also expectedly the adjusted R^2 increases between the CAPM and FF3 regressions. The Adjusted R^2 are high - *the market cap weighted NIFTY100 is a good reflection of the broader market.* More interesting are the loadings of the factors. Both SMB and HML are significant and negatively loaded. As the NIFTY100 comprises of the largest firms in the market, the negative loading is what we would expect. The NIFTY100 constituents are not chosen based on value - just on market cap. The negative loading, the opposite of value, implies that larger firms are more expensive and have higher Market to Book multiples[23]. Finally, and most importantly for our purposes, *one cannot get exposure to momentum simply by taking exposure to the NIFTY100* - the WML coefficient is small and not statistically significant. In addition, FF4 does not improve the R^2 further from FF3. Our choice of universe as the NIFTY100 does not create any systemic bias of momentum.

[22]There are issues in the use of non-U.S factor datasets. See Brückner et al. (2015). While we recognize that caution should be applied, we believe that it is important to use the data sets to explore the use of factor theory in non-US markets.

[23]The long tail of rather illiquid small caps could be the reason for this loading - but we will not formally seek a reason in this paper.

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3.4 Overnight Risk - does our execution strategy make a difference to expected returns

Traditionally, the calculation of factors ignores the fact that all stock exchanges allow trading only for a limited number of hours and generally choose closing prices for the calculations. This has immediate real world issues : an investor cannot use the closing prices to analyse the universe and simultaneously make the investment. Closing prices are the last traded price before the market closes, and the probability of being the last trade is very low. While the skip-month deals with the signal and execution challenge, we believe that a more pragmatic option is to use the closing prices for the portfolio construction, and use open prices for execution[24].

Using closing prices for portfolio construction and opening prices for execution raises the issue of 'overnight' risk : *open prices are not the same as the closing prices of the previous trading day.* With many corporate announcements done after the stock exchange is closed and an increasing number of firms with businesses that are transnational, the 'overnight period'[25] is a source of price jump risk. At the open of a new trading day, the 'overnight' information is reflected in the opening price for the stock. Oldfield and Rogalski (1980) developed the hypothesis that the return distributions for trading and non-trading periods were indeed distinct. Hong and Wang (2000) build a model that demonstrates that '*open-to-open returns are more volatile than close-to-close returns and intraday trading volume is U-shaped*' reflecting empirical evidence. Liu and Tse (2016) show '*overnight returns of US exchange-traded index funds and most international index futures are significantly positive, while returns during trading hours are negative*'. Ahoniemi and Lanne (2013) examine a number of solutions to deal with the volatility of overnight returns.

Open prices are usually determined by the pre-open session where an 'equilibrium' price becomes the official open price. In the event that there is no price discovery in the pre-open session, the price for the first trade in the normal market serves as the opening price. For our analysis, we will use the data from the NIFTY100 index rather than averaging the prices of individual stocks. We do this to illustrate the principle and outline the framework for analysis. An analysis for individual stocks to identify potential stock-specific idiosyncrasies may need to be incorporated in a more detailed implementation.

The NIFTY100 closing prices are calculated by taking the last half an hour weighted average closing prices of the constituents of the index. At open, NIFTY100 opening prices are calculated from the opening prices of the constituents of the index. It is possible, therefore, for stale prices to be part of the calculation of both the open and close prices for the index.

The impact of execution using open prices of the next trading day could be modelled as Equation 8

$$\begin{aligned} r_t^i &= log\left(\frac{Price^i_{open_{\hat{t}}}}{Price^i_{close_{t-1}}}\right) \\ &= log\left(\frac{Price^i_{close_t}}{Price^i_{close_{t-1}}} \times \frac{Price^i_{open_{\hat{t}}}}{Price^i_{close_t}}\right) \\ &= log\left(r^i_{close:t,t-1}\right) + log\left(r^i_{open_{\hat{t}},close_t}\right) \end{aligned} \tag{8}$$

where r_t^i is the log return for stock i at period t, $r^i_{close:t,t-1}$ is the log return for stock i using the

[24] By staying with a liquid large cap universe, we avoid the issue of liquidity constraints and slippages to a large extent.
[25] This includes weekends and public holidays when exchanges are closed

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closing prices between periods t and $t-1$, and $r^i_{open_{\hat{t}},close_t}$ is the log 'return' arising from executing the portfolio actions at the open price on the next trading day ($\hat{t}$) after running the analysis on t. This stylized model will help us in exploring the question 'what is the opportunity loss (or gain) by executing at open on the next trading day'. To answer the question, we first need to explore whether the distribution of daily returns using closing prices is the same as the distribution of returns using opening prices. If the distributions are indeed the same, execution at open would not be systemically different from executing at close. Returns are defined as log returns (Equation 9):

$$r^i_t = log\left(\frac{P^i_t}{P^i_{t-1}}\right) \tag{9}$$

where, r^i_t is the return for stock i for period t and P is the relevant price.

Figures 13 and 14 show the shape of the two distributions and the associated descriptive statistics respectively. The p-value for the Kolmogorov-Smirnov (K-S) test on two samples is 26.30%, indicating that the return distributions are identical. This is important : statistically the underlying distribution of returns between returns using day close and returns using day open are similar. So constructing the portfolio using day close and executing rebalancing/investment at next day open prices should not materially affect the returns of the portfolio just because of the choice of execution price.

Figure 13: Distribution of daily returns using Open, Close Prices

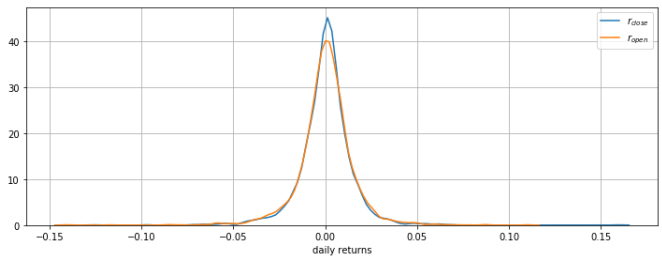

Figure 14: Descriptive Statistics for Distribution of Daily Returns

	Close	Open
min	-0.1264	-0.1408
max	0.1594	0.1100
mean (%)	0.0447	0.0446
std dev (%)	1.3906	1.4709
skewness	-0.1037	-0.4695
kurtosis	10.6211	8.2297

This table shows descriptive statistics of the log daily returns using Close and Open Prices of the NIFTY100.

Next, we turn to the distribution of overnight gap. We define a log-return distribution between the close price (C) of stock i on period t and the open price (O) of i on the next trading day $\hat{t}$ (Equation 10):

$$r^i_{\hat{t}} = log\left(\frac{O_{\hat{t}i}}{C^i_t}\right) \tag{10}$$

By definition, a positive return implies an opportunity loss for the investor since the entry price was higher than the signal price. Figures 15 and 16 show the shape of the distributions and the associated descriptive statistics respectively.

Figure 15: Distribution of overnight returns using Close and Open (next day) Prices

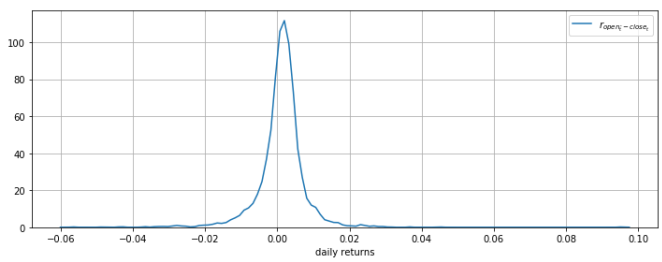

Figure 16: Descriptive Statistics for Distribution of Overnight Returns

	Overnight
min	-0.0579
max	0.0953
mean (%)	0.1140
std dev (%)	0.6618
skewness	-0.3089
kurtosis	21.6555

This table shows descriptive statistics of the log daily returns using Close price on day t and Open Prices on the next trading day of the NIFTY100.

Is the overnight distribution normally distributed? Figure 16 does not seem to indicate so - the

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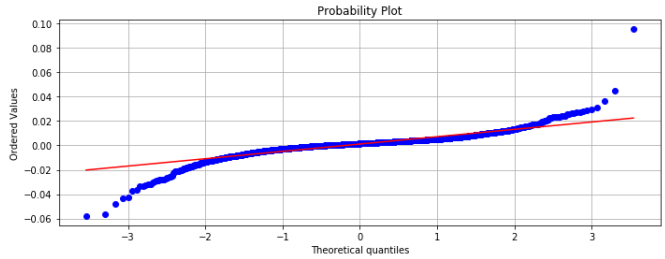
distribution is leptokurtic. This is intuitively what one would expect - news flow overnight will, on occasion, move prices significantly. The quantile on quantile plot (Figure 17) shows this visually.

Figure 17: QQ Plot for returns using Close and Open (next day) Prices



We run a series of statistical tests to check whether the distribution is normal : Shapiro-Wilk, Kolmogorov–Smirnov, Anderson-Darling, and D'Agostino's K^2. All indicate the same outcome : the distribution is not normal (Table 13). The leptokurtic distribution does have a positive mean - there is a bias for open prices to be higher than closing prices. However, an investor can get whipsawed by the occasional large loss. A quantitative strategy using open prices for investment/rebalancing executions is potentially exposed to increased volatility while having a small bias over a theoretical (and impractical) execution strategy that uses close prices as both signal and portfolio execution price.

Table 13: Outcomes of Tests for Normal Distribution

Test	nobs	stat	SigLevel	pvalue	remarks
Shapiro-Wilk	3482	0.83		0.00	Sample does not look Gaussian (reject H0)
Kolmogorov–Smirnov	3482	0.83		0.00	Sample does not look Gaussian (reject H0)
Anderson-Darling	3482	122.23	15.00	0.57	Sample does not look Gaussian (reject H0)
Anderson-Darling	3482	122.23	10.00	0.66	Sample does not look Gaussian (reject H0)
Anderson-Darling	3482	122.23	5.00	0.79	Sample does not look Gaussian (reject H0)
Anderson-Darling	3482	122.23	2.50	0.92	Sample does not look Gaussian (reject H0)
Anderson-Darling	3482	122.23	1.00	1.09	Sample does not look Gaussian (reject H0)
D'Agostino's K^2	3482	944.15		0.00	Sample does not look Gaussian (reject H0)

This table shows summary outcomes of statistical tests to check whether the distribution of the log daily returns using Close price on day t and Open Prices on the next trading day of the NIFTY100 are normally distributed.

Does the overnight risk increase at month-ends? There is evidence of seasonal anomalies (see Ariel (1987), Thaler (1987), Lakonishok and Smidt (1988) amongst others). We run a similar analysis as above for the overnight gaps on the last trading day of a month (Figures 18 and 19).

Figure 18: Distribution of overnight returns using Close and Open (next day) Prices at month-end

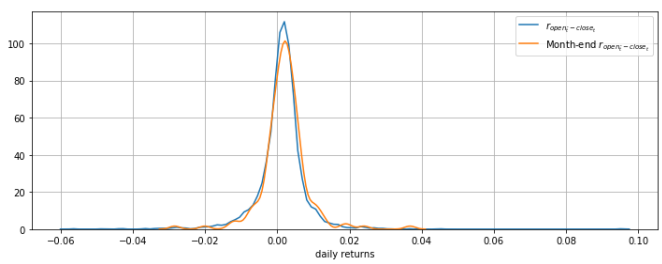


Figure 19: Descriptive Statistics for Distribution of Overnight Returns at month-end

	Overnight
min	-0.0284
max	0.0368
mean (%)	0.2084
std dev (%)	0.6371
skewness	0.3555
kurtosis	8.6297

This table shows descriptive statistics of the log daily returns using Close price on month-end t and Open Prices on the next trading day of the NIFTY100.

This sub-set does seem to have different distribution characteristics than the full sample. The two sample K-S test p-value of 0.2133 statistically rejects the null hypothesis of similar distributions between the overnight gap at month-end with the broader daily overnight gap[26]. Our choice of using

[26]The number of observations is different for the two series. $nobs_{daily} = 3482$ and $nobs_{monthend} = 170$ which gives 0.1068 at the critical value for 5% significance level.

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month-ends does have a specific impact on the returns of the strategy. The series of tests to check for normal distribution also rejects the hypothesis of a normal distribution of returns for the overnight gap at month end (Table 14).

Table 14: Outcomes of Tests for Normal Distribution for overnight risk at month-end

Test	nobs	stat	SigLevel	pvalue	remarks
Shapiro-Wilk	170	0.86		0.00	Sample does not look Gaussian (reject H0)
Kolmogorov–Smirnov	170	0.86		0.00	Sample does not look Gaussian (reject H0)
Anderson-Darling	170	5.57	15.00	0.56	Sample does not look Gaussian (reject H0)
Anderson-Darling	170	5.57	10.00	0.64	Sample does not look Gaussian (reject H0)
Anderson-Darling	170	5.57	5.00	0.77	Sample does not look Gaussian (reject H0)
Anderson-Darling	170	5.57	2.50	0.90	Sample does not look Gaussian (reject H0)
Anderson-Darling	170	5.57	1.00	1.07	Sample does not look Gaussian (reject H0)
D'Agostino's K^2	170	42.33		0.00	Sample does not look Gaussian (reject H0)

This table shows summary outcomes of statistical tests to check whether the distribution of the log daily returns using Close price on day t and Open Prices on the next trading day of the NIFTY100 are normally distributed.

For the specific period under study, we estimate that the month-end overnight risk would make a small but measurable difference to the return of the portfolio. Crudely, the risk bias could translate into an opportunity lost for the investor of around 3 bps per year[27]. Further, the distribution of month-end gap is right-skewed. However, for all practical purposes, the impact of the overnight gap is small. Therefore, the choice of execution prices is both practical and does not make a material difference to the realized returns of an investable momentum strategy. Our analysis also shows that investors can execute systematic strategies using closing prices of one day and execute at open on the next day, subject to capacity. Other solutions like averaging prices for small durations during the trading day are also possible but these give rise to execution uncertainties.

3.5 Is Momentum Persistent?

Having shown that a long only portfolio of large-cap stocks does demonstrate momentum premium, the NIFTY100 does not have an inherent momentum bias and executing rebalancing at the opening price of the first trading day of the following month does not reduce returns in any meaningful manner, we turn to evaluating whether momentum is ***persistent*** i.e. by selecting stocks that have momentum in the recent past, are we successfully selecting stocks that persist in the same direction for the future? Jegadeesh and Titman (1993) had shown that momentum persists in the near term. Beyond 12 months, the performance dissipates. Examining persistence is important to determine possible holding periods.

Our strategy is a top decile of momentum price winners chosen from the NIFTY100 constituents using a 12-month-skip-most-recent month using close prices of the last trading day of a calendar month, executed on the open prices of the first trading day of the following month. To examine the persistence of momentum from this strategy, we build two portfolios every month using our strategy: an equally weighted *Winner* portfolio consisting of constituent stocks with the highest momentum score, and an equally weighted *Loser* portfolio consisting of constituent stocks with the lowest momentum scores for the period. As these portfolios are the top and bottom deciles, we assign a score of 1 (10) for each of the stocks in the winner (loser) portfolio. The mean scores for the portfolios, by construction, are 1 and 10 at inception. We then compute the average mean momentum score for this portfolio from time of the portfolio formation (time t) and in each of the subsequent 12 months (times $t+1$, $t+2$, $t+3$, ... , $t+12$).

Momentum scores by construction vary monotonically between the two portfolios at the time of portfolio formation. We use the mean scores as a proxy for persistence. The average scores trace how the portfolios track momentum over time. If momentum dissipates for our winner portfolio, its mean

[27] the mean overnight gap at month end $\times$ 12.

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score will increase from 1 over time as the individual stocks fall from top decile. If momentum holds, the mean score will stay around 1 as stocks remain in the top decile. Table 15 shows that, on average, and subject to survival of a stock, *momentum does persist in the near term but dissipates over time.*

Table 15: Persistence of Momentum

	Winner Portfolio	Loser Portfolio	W-L	W-L t-stat	W-L p-val
t	1.000	10.000	-9.000		0.000
t+1	1.450	9.609	-8.159	-190.589	0.000
t+2	1.765	9.337	-7.572	-85.146	0.000
t+3	2.064	9.024	-6.960	-51.156	0.000
t+4	2.383	8.661	-6.278	-31.105	0.000
t+5	2.707	8.286	-5.579	-23.380	0.000
t+6	3.025	7.910	-4.885	-16.870	0.000
t+7	3.375	7.505	-4.130	-13.088	0.000
t+8	3.755	7.067	-3.312	-9.286	0.000
t+9	4.103	6.672	-2.569	-6.942	0.000
t+10	4.494	6.299	-1.805	-4.349	0.000
t+11	4.860	5.900	-1.039	-2.332	0.020
t+12	4.997	5.828	-0.831	-1.676	0.094

This table shows average momentum scores. Each calendar month, constituent stocks are ranked in ascending order on the basis of their momentum score. The ranked stocks are assigned to two portfolios : a Winner portfolio consisting of the top ranked stocks by momentum, a Loser portfolio consisting of the bottom ranked stocks. This table reports each portfolio's quality score at portfolio formation (date t) up to the subsequent 12 months (date t + 12 months). We report the time series average of the equal-weighted cross-sectional means. The sample period runs from from Jan 2011 to Sep 2019. Standard errors are adjusted for heteroskedasticity and autocorrelation with a lag length of equivalent to 50% of the period (Newey and West (1987). So at $t + 6$, the lag is 3 and for $t + 12$ it is 6.).

Subject to survival, winner portfolios at portfolio formation keep losing their momentum 'gains' and loser portfolios recover their momentum relatively over the subsequent months. In the near term, the evidence for separation is clear. By the end of the year, however, there is little to separate the original winners and losers. Statistically, the difference by $t + 12$ is not significant at a 5% level.

Our results are similar to Jegadeesh and Titman (1993). We believe that this reflects the behavioural hypothesis of investors over-reacting to information (Bondt and Thaler, 1985, 1987) and their near-term overconfidence (Daniel et al., 1998). The data shows a loss (gain) of one decile for winner (loser) portfolios every passing quarter. This trend is a proxy, we argue, of the evolving reaction of investors to the newsflow over time. In conclusion, **momentum does persist in the near term, and dissipates beyond 12 months**[28] Investors are well advised to review their portfolios frequently so that inherent biases are addressed.

3.6 Analysis of Momentum Returns

We finally turn to examining the returns of our long only monthly rebalanced strategy. As we did in analyzing the NIFTY100, we examine excess returns of our strategy with respect to the CAPM, FF3 and FF4. The alphas are the intercepts from the regression (Equation 11) with the first one, three or four right-hand-side variables included:

$$r_t = \alpha + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{WML} WML + \epsilon_t \tag{11}$$

where r_t is the monthly excess portfolio return over the risk-free rate of the portfolio.

We start by first exploring the difference between the Winner and Loser portfolio. Table 16 shows the results. The results show a sharp difference in outcomes for the two portfolios. The Winner portfolio has significantly higher raw excess excess returns, CAPM and FF3 Alphas. FF4 alphas are not significant for both portfolios. The Winner portfolio has a significantly lower beta and significantly higher Information Ratio and raw sharpe. The Winner Portfolio has a moderate FF4 Adjusted R^2 indicating that FF4 cannot fully explain the excess returns of our Winner portfolio.

[28] For a portfolio with a 1 month hold and a look back period of 12-skip-1 month.

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Table 16: Returns of Momentum : Winner and Loser Portfolios

	Loser Portfolio				Winner Portfolio			
	Co-eff	std err	t-stat	pvalue	Co-eff	std err	t-stat	pvalue
Excess Return	-0.85		-0.99		0.93		2.02	
CAPM Alpha	-1.21	0.47	-2.55	0.01	0.75	0.30	2.50	0.01
FF 3 Factor Alpha	-1.07	0.47	-2.28	0.02	0.80	0.30	2.64	0.01
FF 4 Factor Alpha	0.18	0.40	0.45	0.65	0.20	0.29	0.67	0.51
Beta	1.62				0.80			
Information Ratio	-0.38				0.88			
Raw Sharpe	-0.00				1.08			
FF4 Adjusted R2	0.82			0.00	0.66			0.00

This table shows calendar-time portfolio monthly returns. Each calendar month, the constituents of the NIFTY100 index are ranked in ascending order on the basis of their momentum score. The ranked stocks are assigned to one of two portfolios - a Loser portfolio consisting of the lowest scoring stocks and a Winner portfolio consisting of the highest scoring stocks. Alpha is the intercept in a time series regression of monthly excess return. The explanatory variables are the returns of the market (MKT), size (SMB), book-to-market (HML), and momentum (WML) portfolios from Agarwalla et al. (2013). The sample period runs from Jan 2011 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates. Beta is the realized coefficient on the market portfolio. Raw Sharpe and Information Ratios are monthly. Adjusted R2 is computed for the 4 factor regression.

We next look at the factor loadings for the Winner portfolio. Table 17 summarizes the regression output. First, moving from the one factor, CAPM, to the four factor, FF4, the explanatory power (the adjusted R^2) of the dependent variables increases. The Winner portfolio has a CAPM β (0.80) lower than the broad market or the NIFTY100 index (0.981). For CAPM and FF3, the winner portfolio has a positive (and significant) alpha. Note that the α is moderately higher than the intercept for the WB portfolio against the market. The MKT coefficient is lower for our portfolio than WB - the strategy has less exposure to the market. The lower-than-market beta and a positive alpha are encouraging.

Table 17: Returns of Momentum : Winner Portfolio

	CAPM			FF3			FF4		
	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue	Co-eff	t-stat	pvalue
Intercept	0.75	2.50	0.01	0.80	2.64	0.01	0.20	0.67	0.51
MKT	0.80	11.85	0.00	0.74	9.05	0.00	0.89	11.40	0.00
SMB				-0.01	-0.08	0.94	0.01	0.10	0.92
HML				0.09	1.32	0.19	0.14	2.25	0.03
WML							0.29	5.24	0.00
Adjusted R2	0.58		0.00	0.57		0.00	0.66		0.00

This table shows calendar-time portfolio monthly returns. Each calendar month, the constituents of the NIFTY100 index are ranked in ascending order on the basis of their momentum score. The highest ranked stock decile make our Winner portfolio. Alpha is the intercept in a time series regression of monthly excess return. The explanatory variables are the returns of the market (MKT), size (SMB), book-to-market (HML), and momentum (WML) portfolios from Agarwalla et al. (2013). The sample period runs from Jan 2011 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates.

Factor loadings with FF3 are what we expect. The coefficients for SMB and HML are not significant. Our construction of the portfolio was purely on momentum and did not have size or value selection criteria. And the WML coefficient from the FF4 regression is both positive and statistically significant - a clear indication of the momentum tilt for our long-only portfolio. Recall, our universe (NIFTY100) does not have a momentum bias. Agarwalla et al. (2013) size and momentum portfolios are based on top and bottom 30% of stocks. Our strategy, using a decile cut-off, is a sub-set of their WB portfolio, and has a momentum coefficient of 0.294 - capturing a significant portion of the broader WB portfolio's coefficient to momentum 0.4017.

Our winner portfolio has a positive and significant (at 5%) loading to HML. NIFTY100 had a negative coefficient for this factor. Our winner portfolio seems to be biased towards selecting value stocks within the NIFTY100. We are not sure why this should be the case. Recall, NIFTY100 itself was biased towards growth. In addition, there are studies that show that there is a negative relation between value and momentum (for instance, see Asness et al., 2013; Cakici et al., 2016). Finally, we had expected a bias towards Big in SMB but the realized loading is not significant.

We can examine the relationship between the excess returns of our winner portfolio and the excess

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returns of the WB portfolio from Agarwalla et al. (2013) to determine how much of the WB effect the portfolio is capturing. As we have seen, our NIFTY100 is a subset of the Big (B) portfolio. Equation 12 shows the regression we use to determine the significance of the relationship between the two variables.

$$[r_{winner_t} - RF_t] = \alpha + \beta_{WB}\,[r_{WB_t} - RF_t] + \epsilon_t \tag{12}$$

where r_{winner_t} and r_{WB_t} are the raw returns for the winner portfolio and the WB portfolio for month t respectively and RF_t is the risk-free rate for month t. Table 18 shows the results. The regression has a high R^2 of 0.72 and the winner portfolio captures about 93.35% of WB return effect. This shows that the smaller NIFTY100 portfolio does capture a large part of the WB portfolio's returns. By restricting our universe, we do not lose a lot of the characteristics of the larger universe. Pragmatically, we have simplified the execution mechanics of converting a non-investable factor index into an investable strategy.

Table 18: Winner portfolio and WB portfolio

	Co-eff	t-stat	pvalue
Intercept	0.298	1.204	0.231
WB - RF	0.933	16.256	0.000
Adjusted R2	0.719		0.000

This table shows calendar-time portfolio monthly returns. Each calendar month, the constituents of the NIFTY100 index are ranked in ascending order on the basis of their momentum score. The highest ranked stock decile make our Winner portfolio. Alpha is the intercept in a time series regression of monthly excess return. The explanatory variables are the excess returns of the Winner Big (WB) portfolios from Agarwalla et al. (2013). The sample period runs from Jan 2011 to September 2019. Returns are in INR, do not include currency hedging, and excess returns are over the risk-free rate computed using the 91-days T-bill rate (Agarwalla et al., 2013). Returns and alphas are in monthly percentage, std errors, t-statistics and related p Values are shown next to the coefficient estimates.

Having determined that a systematic long-only winner momentum strategy earns excess returns, has better risk adjusted return characteristics when compared to loser portfolio and that it captures a large component of the WB portfolio's effects, we finally turn to examine performance of our momentum winner portfolio against the NIFTY100. While there are listed NIFTY100 ETFs available for investors, these do not have the historical track record going back to January 2011. So we use the raw NIFTY100 index prices as our comparison benchmark.

We compute price returns on an open-to-open basis. We ignore trading and custody costs and any taxes - in the current context, these costs are fairly small (many discount brokers today offer 'brokerage-free' trading of stocks)[29] and do not impair the overall results. We ignore dividends and, therefore, total returns in our calculations.

Some comments on the costs of implementing such a strategy to determine whether there will be a significant deterioration after costs are in order. Total trading costs include brokerage, applicable taxes and potential bid-ask spreads. With the growth of strong discount brokers in India, general investors are able to execute trades at low or no brokerage fees. India has a range of small charges including taxes (Securities Transaction Tax 0.1% on purchase and sale and Goods and Service Tax at 18% on brokerage and transaction charges), transaction charges by the stock exchanges, SEBI Turnover Charges (0.00015% of the transaction value) and some taxes levied by state governments. These costs are slightly in excess of 0.1 bps per leg. Using a discount broker with free/low brokerage, then, will not reduce the performance materially. Discount brokers, generally, have better technology platforms than traditional brokers making it more convenient for investors to place overnight orders.

Figures 20, 21 and 22 summarise the performance of the momentum portfolio versus the NIFTY100

[29] Outside, the brokerage costs, other associated costs and taxes are less than 5 bps. In comparison, ETFs tracking the NIFTY100 have a Total Expense Ratios between 26 and 48 bps. The cost of running a 10 stock portfolio with very high monthly turnover should be comparable to these costs under the current discount brokerage cost structure.

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since 1 January 2011.

Figure 20: Systematic IN Momentum Strategy vs NIFTY100 Monthly Price Return Analysis Month Start to Month Start

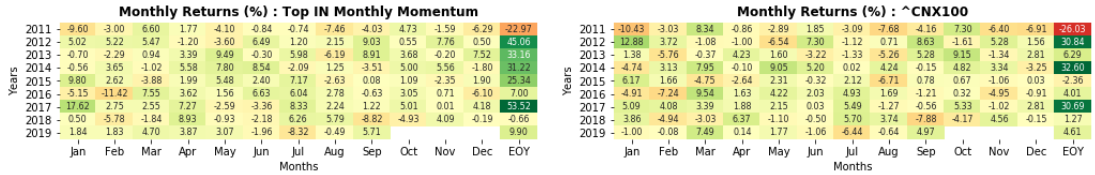

Figure 21: Systematic IN Momentum Strategy Return Analysis: Month Start to Month Start

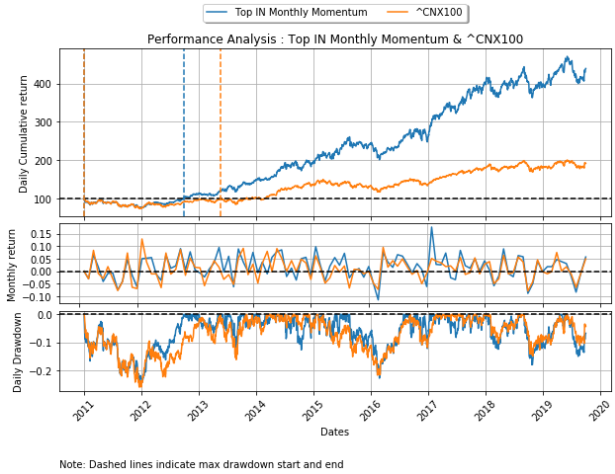

Figure 22: Systematic IN Momentum Strategy Summary Statistics

Stat	Top IN Monthly Momentum	^CNX100
Start	2011-01-03	2011-01-03
End	2019-09-30	2019-09-30
Risk-free rate	0.00%	0.00%
Total Return	338.90%	91.81%
CAGR	18.44%	7.74%
Max Drawdown	-24.59%	-26.12%
1Y	8.76%	3.76%
3Y (ann.)	18.18%	9.83%
5Y (ann.)	19.55%	8.10%
Since Incep. (ann.)	18.44%	7.74%
Daily Sharpe	1.02	0.55
Daily Sortino	1.68	0.93
Daily Info Ratio	0.48	-
Calmar Ratio	0.75	0.30
Daily Mean (ann.)	19.20%	8.99%
Daily Vol (ann.)	18.84%	16.25%
Best Day	5.39%	7.35%
Worst Day	-7.77%	-5.59%
Monthly Sharpe	1.08	0.55
Monthly Sortino	2.35	1.14
Monthly Info Ratio	0.88	-
Monthly Mean (ann.)	18.44%	8.67%
Monthly Vol (ann.)	17.01%	15.62%
Best Month	17.62%	12.88%
Worst Month	-11.42%	-10.43%
Yearly Sharpe	0.84	0.47
Yearly Sortino	2.64	1.06
Yearly Info Ratio	0.93	-
Yearly Mean	20.17%	9.10%
Yearly Vol	24.04%	19.27%
Best Year	53.52%	32.60%
Worst Year	-22.97%	-26.03%
Avg. Drawdown	-3.19%	-3.03%
Avg. Drawdown Days	28.73	46.08
Avg. Up Month	4.55%	4.05%
Avg. Down Month	-3.37%	-3.08%

The long term CAGR of 18.44% pa for our momentum strategy versus 7.74% pa for the NIFTY100 over the period, a spread of over 10.70% pa, indicates that investing in a systematic momentum strategy does payoff over the long term. Monthly raw sharpe's of the portfolio (1.08) is 1.95x the raw sharpe of the NIFTY100 (0.55), demonstrating superior risk adjusted return characteristics of the momentum portfolio. *A priori*, we expected that our momentum portfolio will have higher volatility than the NIFTY100. Overall monthly volatility for the momentum strategy (17.01%) is indeed higher than the monthly volatility of the broader NIFTY100 index (15.62%).

Rather than "point-in-time" results anchored by the end of a calendar period like a month or year, rolling returns account for the fact that investors can invest at any point in time and typically do not invest at the beginning of the current one-, three- or five-year period. Rolling returns are returns taken over a continuous period of time : a particular period of returns, say 1 year, is taken every day over the period of time under analysis. This gives an indication of the range of outcomes for an investor across a period of time. Rolling returns provide a robust analytical tool for evaluating performance. This is especially important during volatile periods when entering a strategy a few months forward or backward from any given date can deliver vastly different returns. We show rolling returns in Figure 23. We use a standard box plot to show the variation (max, min, mean, inter quartile range) over 1, 3 and 5 year periods. We compare the rolling returns of the strategy with those of NIFTY100.

We flag a few points. First, momentum strategy returns are more volatile in an absolute sense than the index returns in the short term. It is, therefore, not a strategy for investors who do not have adequate risk appetite for volatility. Over longer periods, the variation of performance shrinks. Second, over time, while short run horizons do show the impact of *momentum crashes*, the data shows

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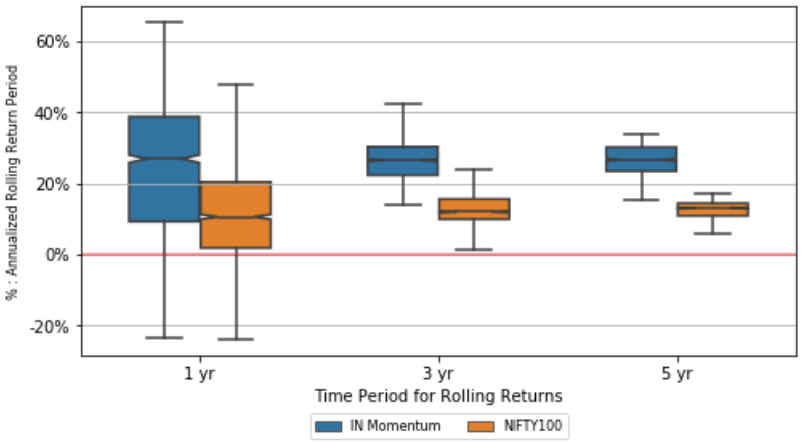
Figure 23: IN Momentum Strategy & NIFTY100 Rolling Returns



that recovery from crashes is fairly quick. Third, the out-performance of momentum as a strategy is more evident the longer one is invested. This reflects the old market adage : *time in the market is better than timing the market.* Momentum investors need to have conviction and stay the course despite short-term volatility.

We finally look at portfolio turnover. The turnover of a portfolio is an important consideration for a retail investor. This is true despite the low/no brokerage costs for execution. The administrative overhead of managing a high turnover portfolio can be onerous. Our portfolio is rebalanced monthly: *a priori*, momentum should have a high turnover. The profile of the rolling monthly turnover[30] of the Winner portfolio is shown in Figure 24 (with a max of 60.00%, a min of 0.00%, a mean of 32.10% and a mode of 20.00%). This turnover profile is higher than reported by say Babbar and Sehgal (2018) (who report an average annual turnover of 89% in their study of 237 mutual funds in India between 2007 and March 2013) or Israel et al. (2017b) (who report annualized one sided turnover numbers between 79% (US Small Cap) and 89.9% (International) for AQR Momentum Strategies between July 2009 and December 2016.). These studies have looked at well diversified portfolios with large number of stocks. Our portfolio has 10 stocks and therefore, should report higher turnover rates relative to portfolios with larger number of stock. Portfolio turnover can be reduced through a number of strategies, including lengthening holding periods, having range bands of momentum scores before stocks are dropped from the portfolio. We do not explore these in this paper as we believe that the portfolio turnover is administratively manageable for an investor and the low implementation costs will not detract from the strategy's performance.

4 Conclusion

We show that the momentum anomaly is strong in the Indian equity market and analyse the components of momentum. In general, Winners contribute more than Losers and $Winner_{Small}$ contributes significantly to the momentum premium. $Winner_{Big}$ contributes steadily to WML. In more recent history, $Winner_{Big}$ has performed than $Winner_{Small}$. *Loser* portfolios, irrespective of size, contribute only occasionally during periods of broader economic stress. In general, short portfolios have been net detractors to momentum performance. Momentum is subject to occasional, significant momentum crashes, both in the Big and Small size portfolios and generally, more for Loser portfolios than Winner portfolios. We show that momentum has persistence in the near term which dissipates over

[30] We calculate the turnover over previous month. We believe this is the most representative for an investor measuring turnover for portfolios rebalanced monthly.

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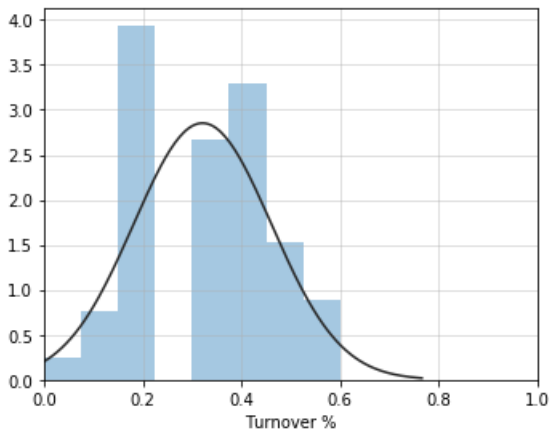
Figure 24: Turnover of Winner portfolio



12 months. This requires investors who want to invest in momentum to do near-term rebalancing but stay invested in the momentum factor over the long term.

Given the restriction (prohibition) of shorting stocks in the Indian market, and the potential for degradation of theoretical returns from Small companies arising from liquidity and capacity, we construct a systematic, monthly-rebalanced, long-only momentum portfolio using 'off-the-shelf' criteria based on the most liquid NIFTY100 constituents and show that such a strategy outperforms the NIFTY100 on both nominal and risk adjusted returns. Such a portfolio also captures a large portion of the returns from $Winner_{Big}$ portfolio despite having a smaller number of stocks in both the universe and the portfolio. We show a practical, easy-to-adopt implementation strategy which gives general investors exposure to the momentum tilt. The costs of such a strategy are also nominal - and the out performance of the momentum tilt does survive real-world implementation. We demonstrate that the results are robust over time. In the absence of liquid ETFs in the Indian market, general investors looking for exposure to the momentum factor would be able to construct low cost, administratively easy to manage factor-tilt long only portfolios.

A logical extension of this work would be to explore the inclusion of a sub-set of *Small* size firms to the portfolio subject to the associated capacity/liquidity and slippage risks. A more formal estimation of capacity and liquidity related costs is another area to be explored.

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